



Practical Report - <08>
SE3082 - Parallel Computing
3rd year 1st semester - 2026

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Cluster setup on AWS. Two t3.micro EC2 instances, Master and Worker, in the same VPC and security group, with passwordless SSH from Master to Worker and an NFS-shared /shared folder holding the binaries and the machinefile. The hostname run confirms all four ranks are distributed two per node.

```
ubuntu@ip-172-31-17-159:/shared$ cat /shared/machinefile
mpirun --hostfile /shared/machinefile --map-by node -np 4 hostname
172.31.17.159 slots=2
172.31.19.90 slots=2
ip-172-31-17-159
ip-172-31-17-159
ip-172-31-19-90
ip-172-31-19-90
ubuntu@ip-172-31-17-159:/shared$ |
```

Exercise 2

The range 1 to 10,000,000 is split into equal blocks across the ranks. Each rank sums its own block, then sends the partial sum to rank 0 with MPI_Send while rank 0 collects them with MPI_Recv. The result 50,000,005,000,000 matches the closed form $N(N+1)/2$, confirming no values were lost or double counted.

```
ubuntu@ip-172-31-17-159:/shared$ mpirun --hostfile /shared/machinefile --map-by node -np 4 /shared/ex02_sum
rank 2 of 4 : range [5000001, 7500000] local_sum = 15625001250000 (0.001286 s)
rank 0 of 4 : range [1, 2500000] local_sum = 3125001250000 (0.003449 s)
rank 3 of 4 : range [7500001, 10000000] local_sum = 21875001250000 (0.003072 s)
rank 1 of 4 : range [2500001, 5000000] local_sum = 9375001250000 (0.001239 s)

N                = 10000000
processes        = 4
parallel sum     = 50000005000000
expected sum     = 50000005000000 (MATCH)
time             = 0.003449 s
CSV,4,0.003449
ubuntu@ip-172-31-17-159:/shared$
```

Exercise 3

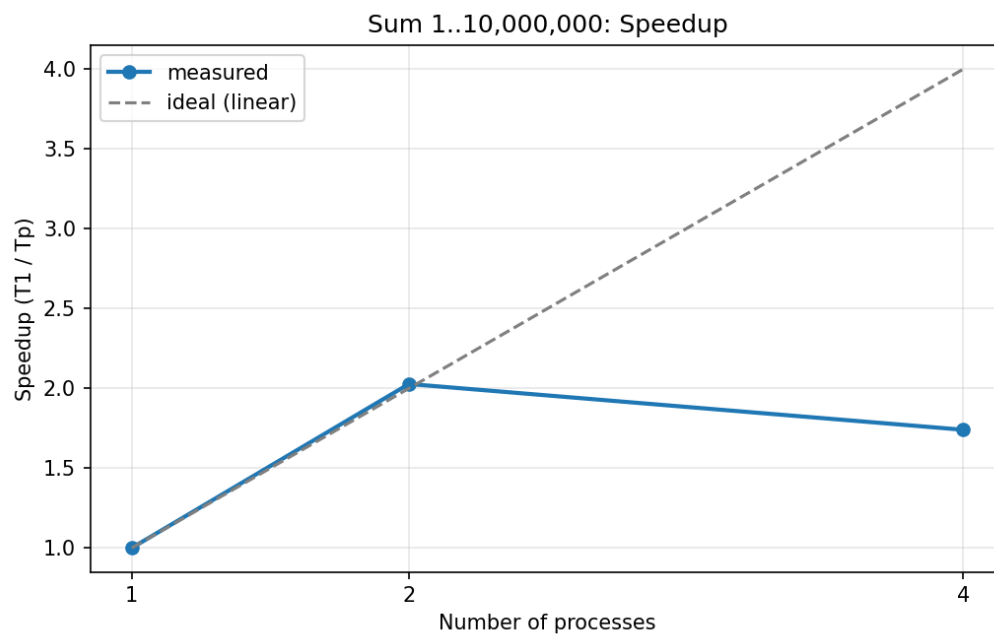
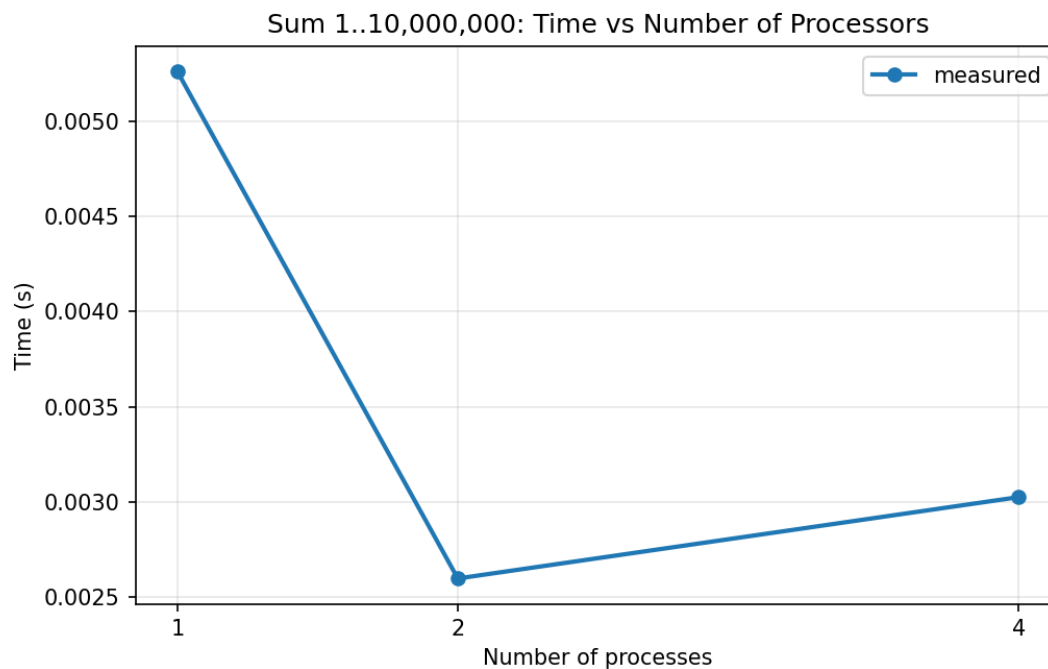
Pi is estimated by the Monte Carlo method over 10,000,000 darts, divided evenly across the ranks. Each rank uses its own seed so the ranks do not repeat the same random sequence, counts the darts falling inside the quarter circle, and sends its hit count to rank 0. Rank 0 sums the hits and computes $4 \times \text{hits} / \text{darts}$.

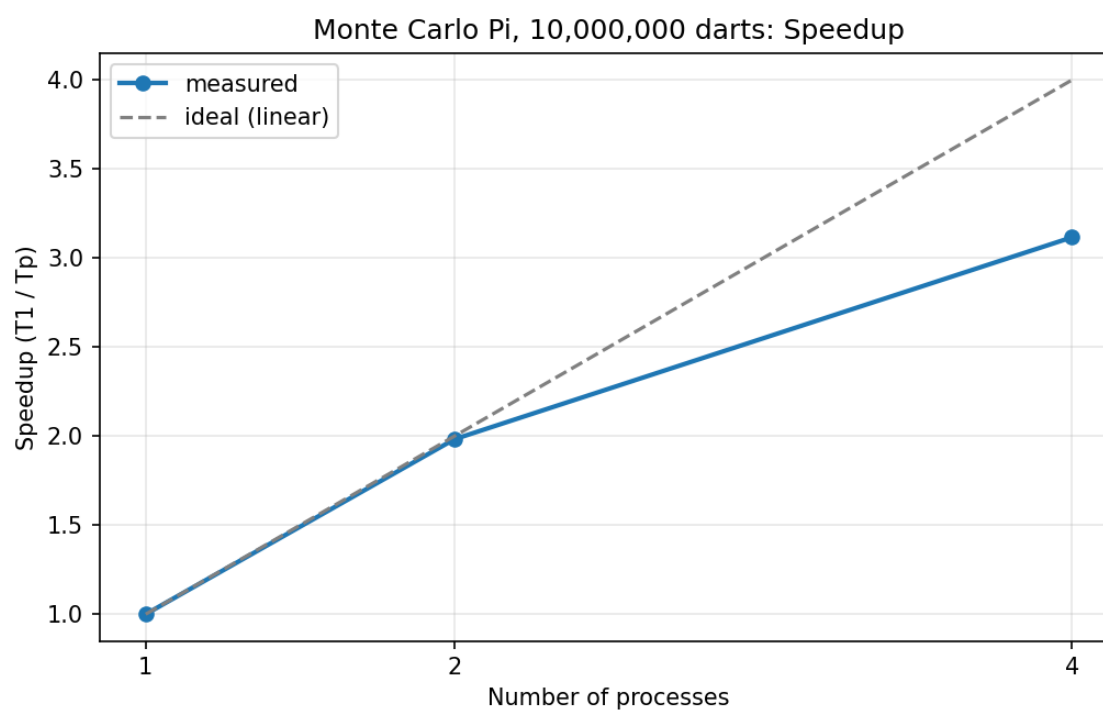
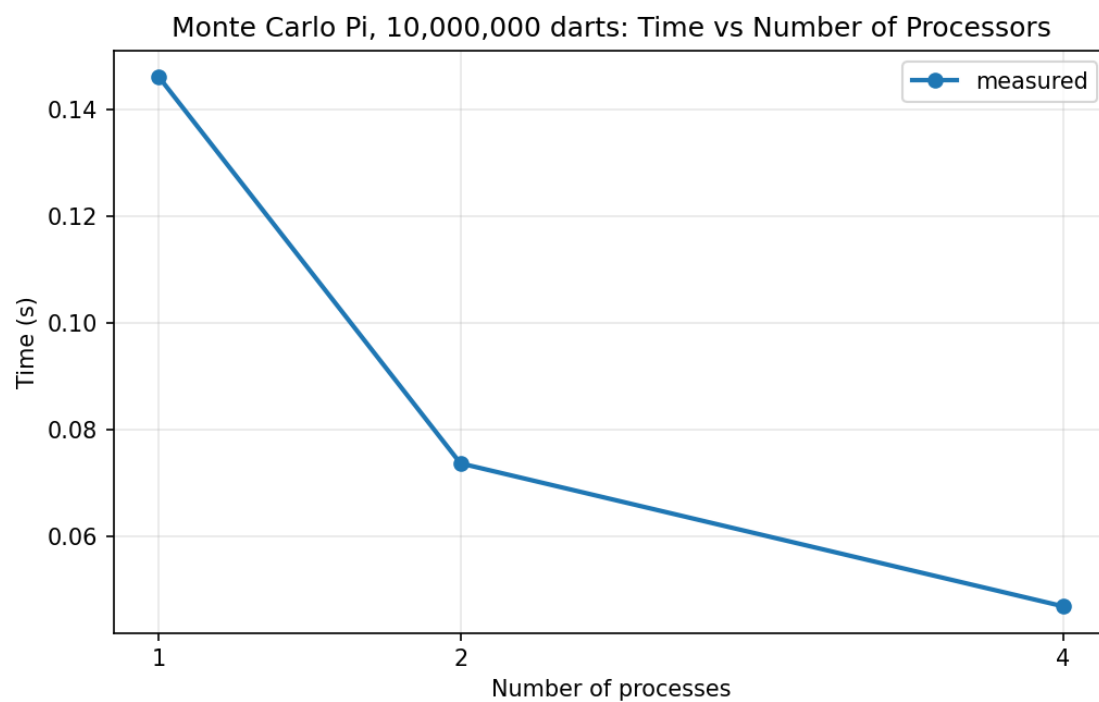
```
ubuntu@ip-172-31-17-159:/shared$ mpirun --hostfile /shared/machinefile --map-by node -np 4 /shared/ex03_pi
rank 2 of 4 : darts = 2500000, hits = 1964183 (0.120738 s)
rank 0 of 4 : darts = 2500000, hits = 1962461 (0.120782 s)

darts           = 10000000
processes        = 4
hits             = 7855103
pi estimate      = 3.1420412000
error            = 0.0004485464
time             = 0.120782 s
CSV,4,0.120782
rank 3 of 4 : darts = 2500000, hits = 1964043 (0.120387 s)
rank 1 of 4 : darts = 2500000, hits = 1964416 (0.061957 s)
ubuntu@ip-172-31-17-159:/shared$ |
```

Exercise 4

Both programs were timed at 1, 2 and 4 processes, repeated five times with the best time kept, and speedup calculated as T_1 / T_p . Speedup rises from 1 to 2 processes and then flattens, because each t3.micro exposes 2 vCPUs on a single physical core, so the two-node cluster provides only about two cores of real compute and the remaining ranks contend for the same hardware.





Exercise 5

Part 1 shows what happens when the source and destination do not match. Rank 0 still reports that MPI_Send returned, because the 4 byte message is small enough to be copied into internal buffers under the eager protocol without a matching receive. Rank 1 blocks inside MPI_Recv forever and MPI reports no error, since an unmatched receive stays legal until the job ends, so the program simply deadlocks until it is killed. A destination rank outside the communicator is different: MPI detects it locally and aborts at once with MPI_ERR_RANK. Part 2 rewrites message2 with MPI_Bsend, attaching a user buffer sized for three messages and giving each send its own variable so no send buffer is overwritten while a message is still outstanding.

```
ubuntu@ip-172-31-17-159:/shared$ timeout 15 mpirun --hostfile /shared/machinefile --map-by node -np 2 /shared/ex05_1_mismatch 2
rank 0: sending 42 to rank 1 with tag 0
rank 1: waiting for a message from rank 0 with tag 99
rank 0: MPI_Send returned
[ip-172-31-19-90:00000] *** An error occurred in Socket closed
[ip-172-31-19-90:00000] *** reported by process [759627777,1]
[ip-172-31-19-90:00000] *** on a NULL communicator
[ip-172-31-19-90:00000] *** Unknown error
[ip-172-31-19-90:00000] *** MPI_ERRORS_ARE_FATAL (processes in this communicator will now abort,
[ip-172-31-19-90:00000] *** and MPI will try to terminate your MPI job as well)
-----
Sorry! You were supposed to get help about:
  prun:proc-ordered-abort
But I couldn't open the help file:
  /usr/lib/x86_64-linux-gnu/pmix2/share/pmix/help-prun.txt: No such file or directory
  /usr/lib/x86_64-linux-gnu/prte3/share/prte/help-prun.txt: No such file or directory.
Sorry!
-----
ubuntu@ip-172-31-17-159:/shared$
```

```
ubuntu@ip-172-31-17-159:/shared$ mpirun --hostfile /shared/machinefile --map-by node -np 2 /shared/ex05_1_mismatch 3
rank 0: sending 42 to rank 2 with tag 0
rank 1: waiting for a message from rank 0 with tag 0
[ip-172-31-17-159:00000] *** An error occurred in MPI_Send
[ip-172-31-17-159:00000] *** reported by process [61341697,0]
[ip-172-31-17-159:00000] *** on communicator MPI_COMM_WORLD
[ip-172-31-17-159:00000] *** MPI_ERR_RANK: invalid rank
[ip-172-31-17-159:00000] *** MPI_ERRORS_ARE_FATAL (processes in this communicator will now abort,
[ip-172-31-17-159:00000] *** and MPI will try to terminate your MPI job as well)
-----
Sorry! You were supposed to get help about:
  prun:proc-ordered-abort
But I couldn't open the help file:
  /usr/lib/x86_64-linux-gnu/pmix2/share/pmix/help-prun.txt: No such file or directory
  /usr/lib/x86_64-linux-gnu/prte3/share/prte/help-prun.txt: No such file or directory.
Sorry!
-----
ubuntu@ip-172-31-17-159:/shared$ |
```

```
ubuntu@ip-172-31-17-159:/shared$ mpirun --hostfile /shared/machinefile --map-by node -np 2 /shared/ex05_2_bsend
Process 0 sent 0 (buffered)
Process 0 sent 10 (buffered)
Process 0 sent 20 (buffered)
Process 1 received 0
Process 1 received 10
Process 1 received 20
ubuntu@ip-172-31-17-159:/shared$
```

Exercise 6

The receive in Exercise 3 is changed to accept `MPI_ANY_SOURCE`. Note that `MPI_ANY_SOURCE` is the wildcard for the source argument of `MPI_Recv` and not for the tag, whose wildcard is `MPI_ANY_TAG`. Rank 0 now takes results in arrival order rather than fixed rank order and uses `status.MPI_SOURCE` to identify each sender. The Pi estimate is identical to Exercise 3 because the hit counts are integers and integer addition is associative, so the order of accumulation cannot change the total. Only the arrival order varies between runs, and with balanced work the timing difference is within noise.

```
ubuntu@ip-172-31-17-159:/shared$ mpirun --hostfile /shared/machinefile --map-by node -np 4 /shared/ex06_pi_any
rank 0: got 1964416 hits from rank 1 (tag 0)
rank 0: got 1964183 hits from rank 2 (tag 0)
rank 0: got 1964043 hits from rank 3 (tag 0)
rank 2 of 4 : darts = 2500000, hits = 1964183 (0.045555 s)
rank 0 of 4 : darts = 2500000, hits = 1962461 (0.046870 s)

[MPI_ANY_SOURCE version]
darts      = 10000000
processes  = 4
hits       = 7855103
pi estimate = 3.1420412000
error      = 0.0004485464
time       = 0.046870 s
CSV,4,0.046870
rank 3 of 4 : darts = 2500000, hits = 1964043 (0.046568 s)
rank 1 of 4 : darts = 2500000, hits = 1964416 (0.044951 s)
ubuntu@ip-172-31-17-159:/shared$
```

Exercise 7

Exercise 6 repeated with the workers using Buffered Send. Each worker attaches its own buffer, calls `MPI_Bsend` and returns immediately without waiting for rank 0 to post the matching receive, then detaches before finalising so the message is guaranteed to have left. The results match Exercise 6 exactly; what changes is that the send is local and decoupled from the receiver's progress.

```
ubuntu@ip-172-31-17-159:/shared$ mpirun --hostfile /shared/machinefile --map-by node -np 4 /shared/ex07_pi_any_bsend
rank 0: got 1964416 hits from rank 1 (buffered send)
rank 0: got 1964183 hits from rank 2 (buffered send)
rank 0: got 1964043 hits from rank 3 (buffered send)
rank 2 of 4 : darts = 2500000, hits = 1964183 (0.045294 s)
rank 0 of 4 : darts = 2500000, hits = 1962461 (0.046974 s)
rank 3 of 4 : darts = 2500000, hits = 1964043 (0.046552 s)
rank 1 of 4 : darts = 2500000, hits = 1964416 (0.044950 s)

[MPI_ANY_SOURCE + MPI_Bsend version]
darts      = 10000000
processes  = 4
hits       = 7855103
pi estimate = 3.1420412000
error      = 0.0004485464
time       = 0.046974 s
CSV,4,0.046974
ubuntu@ip-172-31-17-159:/shared$
```